Angles of Elevation & Depression

Exam code: 0607

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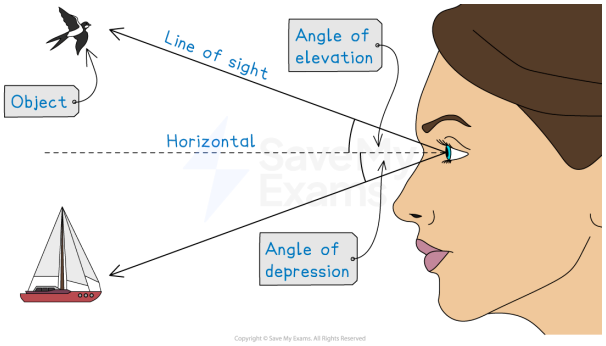
Elevation & depression

What are angles of elevation and depression?

- An **angle of elevation** or **depression** is the angle measured between the **horizontal** and the **line of sight**
  - Looking **up** at an object creates an angle of **elevation**
  - Looking **down** at an object creates an angle of **depression**
- **Right-angled trigonometry** can be used to find
  - an **angle** of elevation or depression
  - or a missing **distance**
- The **tan ratio** is often used in real-life scenarios
  - You may **know the height** of an object and want to **find the distance** you are from it
  - You may **know the distance** you are from an object and want to **find its height**



Help me understand

Examiner Tips and Tricks

It may be useful to draw more than one diagram if the triangles that you are interested in overlap one another.

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

A cliff is perpendicular to the sea and the top of the cliff, *T*, stands 24 metres above the level of the sea.

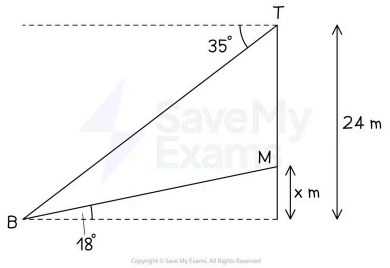
The angle of depression from the top of the cliff to a boat at sea is 35°.

At a point x metres vertically up from the foot the cliff is a flag marker, *M*.

The angle of elevation from the boat, *B*, to the flag marker is 18°.

(a) Draw a diagram of the situation. Label all the angles and distances given above.

Answer:

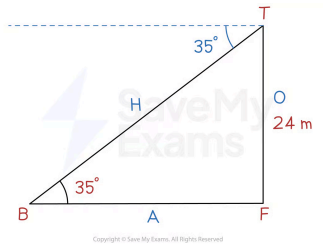


(b) Find the distance from the boat to the foot of the cliff.

Answer:

Consider triangle *TBF* where *F* is the foot of the cliff
Angle *TBF* = 35° because of alternate angles

Use SOHCAHTOA to find the missing distance
We know the opposite (*TF*) and we want to find the adjacent (*BF*), so use $\tan\theta = \frac{O}{A}$



$$\tan 35 = \frac{24}{BF}$$

$$BF = \frac{24}{\tan 35}$$

$$BF = 34.27555...$$

$$BF = 34.3 \text{ m (3 s.f.)}$$

(c) Find the value of x.

Answer:

Consider triangle *MBF*

A

$$\tan 18 = \frac{x}{34.27555...}$$

$$34.27555... \tan 18 = x$$

$$x = 11.1368...$$

$$x = 11.1 \text{ m } (3 \text{ s.f.})$$

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